

2017-08-29 TRAFFIC ANALYSIS POP QUIZ - INCIDENT REPORT

EXECUTIVE SUMMARY:

On Monday 2017-08-28, an item of malicious spam (malspam) with the subject line "fattura agosto" was discovered. The email has a Microsoft Excel spreadsheet as an attachment. We tested the attachment on 2017-08-29, and found it has a malicious macro that, if enabled on a vulnerable Windows host, will infect it with the [Chthonic banking Trojan](#). Details follow.

EMAIL INFORMATION:

- Date: Monday, 2017-08-28 11:28 UTC
- From: "Stagio Basile" <maperio1@fastwebnet.it>
- Subject: fattura agosto
- Attachment: 5217 FATTURA-SALDO.xls

ASSOCIATED MALWARE:

SHA256 hash for attached Excel spreadsheet:

- 69542def1cf59f5a61dd16ad415445ef2daeb5c519fcbe89bb969106dbf01b90
 - [VirusTotal link](#)
 - [Payload Security sandbox analysis](#)

SHA256 hash for follow-up malware downloaded by the Excel macro:

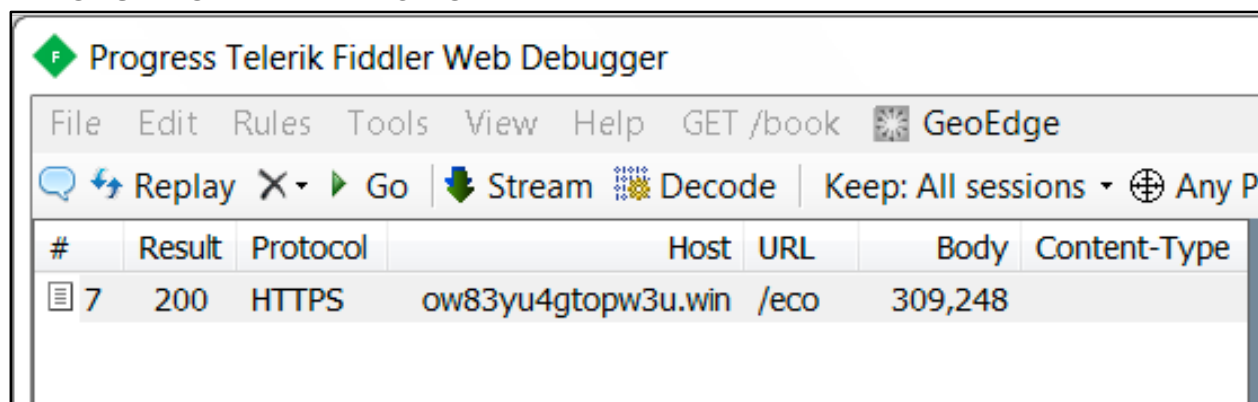
- c81ea524d410ba9ab49524d0c446075a0f0112f073ef973d328f69199371c08d
 - [VirusTotal link](#)
 - [Payload Security sandbox analysis](#)
 - Location: C:\Users\[username]\AppData\Roaming\WindowsPhotoViewery\WindowsPhotoViewery.exe

NETWORK TRAFFIC INDICATORS:

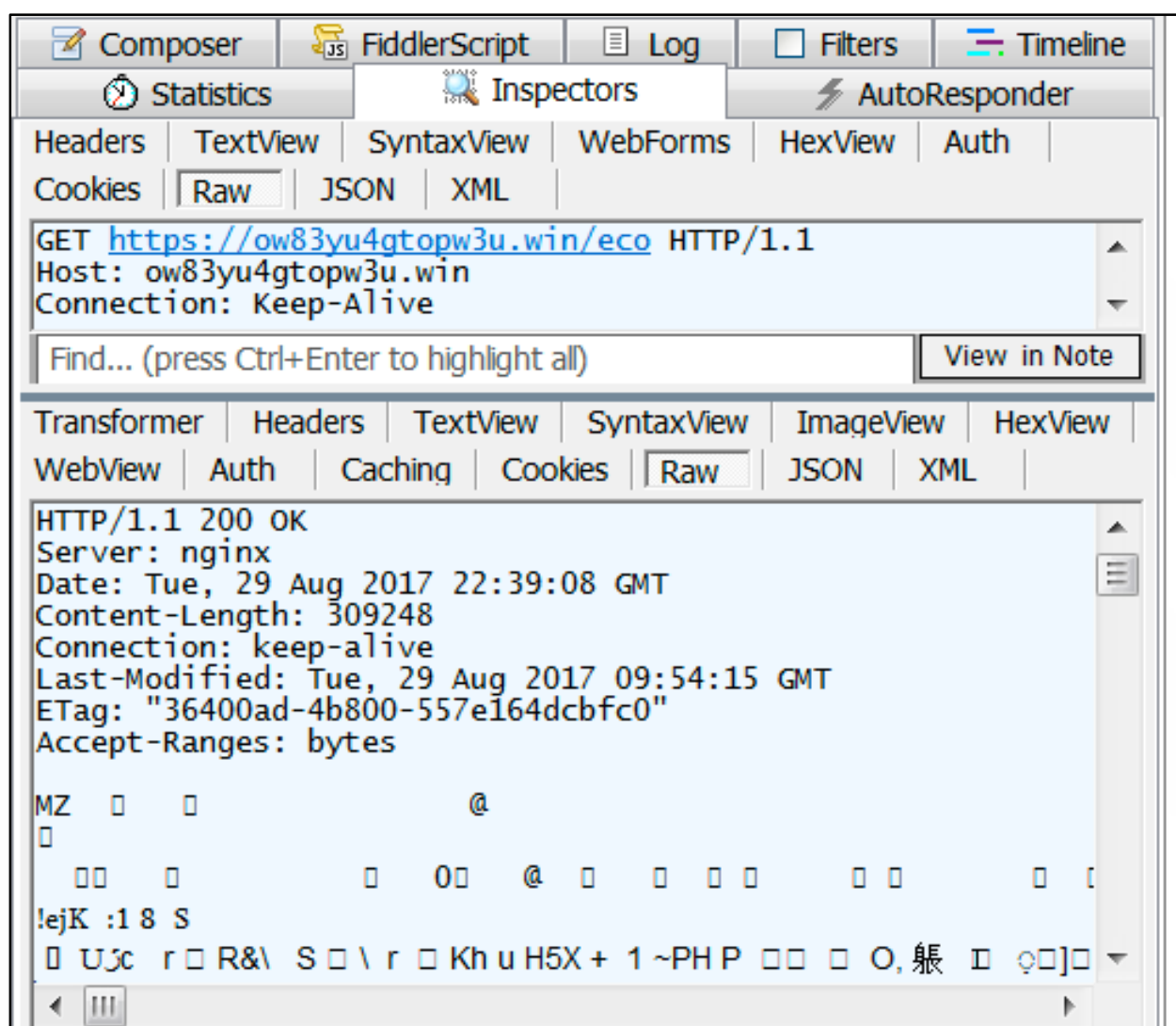
- 119.28.47.202 port 443 - **ow83yu4gtopw3u.win** - GET /eco (HTTPS, downloaded Chthonic)
- 119.28.15.177 port 80 - **amellet.bit** - POST /a/
- 119.28.15.177 port 80 - **amellet.bit** - POST /html/
- 23.94.5.133 TCP port 53 - dns queries for amellet.bit
- 37.187.16.17 TCP port 53 - dns queries for amellet.bit
- 51.255.48.78 TCP port 53 - dns queries for amellet.bit
- 45.56.117.118 TCP port 53 - attempted connection, no response from server
- 45.63.99.180 TCP port 53 - attempted connection, no response from server
- 85.159.213.210 TCP port 53 - attempted connection, no response from server
- 108.61.164.218 TCP port 53 - attempted connection, no response from server

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IMAGES FROM THE INFECTION:



Shown above: HTTPS URL generated by macro from the attached Excel spreadsheet.



Shown above: More information on HTTPS traffic caused by the Excel macro.

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Filter: `ssl.handshake.type == 1 or http.request or dns.qry.nan` Expression... Clear Apply Save Filter Filter Filter

Date/Time	Dst	port	Host	Info
2017-08-29 22:25:17		53		Standard query 0x429b A ow83yu4gtopw3u.win
2017-08-29 22:25:17				Standard query response 0x429b A 119.28.47.202
2017-08-29 22:25:18	119.28.47.202	443		Client Hello
2017-08-29 22:26:24	119.28.15.177	80	amellet.bit	POST /a/ HTTP/1.0
2017-08-29 22:27:12	119.28.15.177	80	amellet.bit	POST /html/ HTTP/1.0
2017-08-29 22:27:27	119.28.15.177	80	amellet.bit	POST /a/ HTTP/1.0
2017-08-29 22:28:19	119.28.15.177	80	amellet.bit	POST /html/ HTTP/1.0
2017-08-29 22:29:08	119.28.15.177	80	amellet.bit	POST /html/ HTTP/1.0
2017-08-29 22:29:12	119.28.15.177	80	amellet.bit	POST /html/ HTTP/1.0
2017-08-29 22:29:16	119.28.15.177	80	amellet.bit	POST /html/ HTTP/1.0
2017-08-29 22:29:18	119.28.15.177	80	amellet.bit	POST /html/ HTTP/1.0
2017-08-29 22:29:21	119.28.15.177	80	amellet.bit	POST /html/ HTTP/1.0

Shown above: Traffic from the lab infection filtered in Wireshark.

RealTime Events Escalated Events

ST	CNT	Date/Time	Src IP	SPort	Dst IP	DPort	Pr	Event Message
RT	3	2017-08-29...	119.28.47.202	443	10.8.29.102	49165	6	ET POLICY Lets Encrypt Free SSL Cert Observed
RT	4	2017-08-29...	10.8.29.102	51797	37.187.16.17	53	6	ETPRO TROJAN Possible Chthonic DNS Lookup
RT	2	2017-08-29...	10.8.29.102	51798	119.28.15.177	80	6	ETPRO TROJAN Chthonic CnC Beacon 9
RT	19	2017-08-29...	10.8.29.102	63896	119.28.15.177	80	6	ETPRO TROJAN Chthonic CnC Beacon 8

Shown above: Alerts from the network traffic in Security Onion using Suricata.

Registry Editor

Name	Type	Data
(Default)	REG_SZ	(value not set)
WindowsPhotoViewery	REG_SZ	C:\Users\[username]\AppData\Roaming\WindowsPhotoViewery\WindowsPhotoViewery.exe

Computer\HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run

WindowsPhotoViewery.exe

WindowsPhotoViewery.exe Properties

Property	Value
Description	
File description	Empty Faster Thirteen
Type	Application
File version	2.3.8.741
Product name	ConsoleAffects
Product version	2.3.8.741
Copyright	Copyright © 2000 - 2014 KG and its Licensors ...
Size	302 KB
Date modified	8/29/2017 10:25 PM
Language	Danish (Denmark)
Legal trademarks	Copyright © 2000 - 2014 KG and its Licensors ...
Original filename	ConsoleAffects.exe

Size: 302 KB (309,248 bytes)

Remove Properties and Personal Information

OK Cancel Apply

Shown above: Chthonic malware made persistent on the infected Windows host.